

Inverse Butterfly 1-2-1 (SPX) — Verdict

honest conclusion · validation stage

QuantConnect · 2021-2026 · full grid: 6 DTE x 6 widths x exit-rules · hourly

WHAT IT IS — short call fly (+2 ATM / -1 ATM-W / -1 ATM+W), net credit, long-vol.

Profits on movement; max gain = the credit; defined risk.

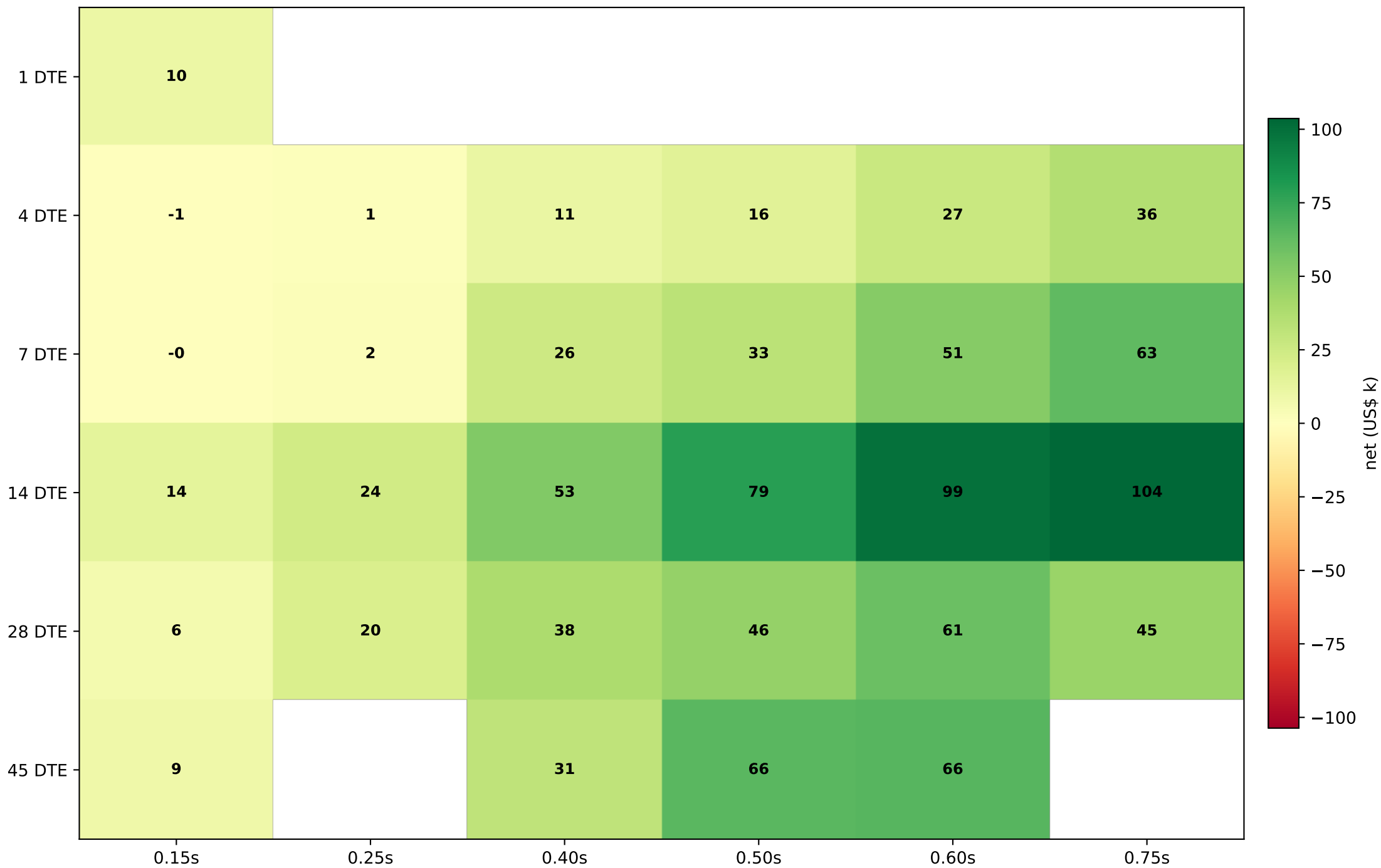
VERDICT — not a convincing standalone edge.

- Best config: 14 DTE @ 0.75 sigma (hold) = \$+103,659 mid -> ~\$+66,459 after slippage, win rate 63%. That's the CEILING and it's modest (~\$12k/yr).
- The winners are HOLD with WIDE wings (0.60-0.75 sigma); TP and early exits do NOT help.
- Structural headwind: implied vol > realized (~0.73-0.85) — you overpay for movement.
- P&L is lumpy (few big-move months carry it); mid is optimistic.

RECOMMENDATION: not a priority / not forward-test ready as a standalone strategy. Possible future use only as a small long-vol DIVERSIFIER (pays when the short-vol book bleeds).

Note: TP figures were corrected for a stale-quote artifact (see methodology page).

1. Hold net P&L heatmap — DTE x width (US\$ k, mid)



= more profit. Edge concentrates in WIDE wings at medium DTE (14) — but it's mid pricing and lumpy. Narrow wings (0.15s) and short DTE (1/4/7) are weak. Best cells = hold, no

2. Top configs — mid vs realistic (hold)

realistic = mid minus ~\$150/trade slippage (verified at minute) · 5.5 years

Config	Trades	Net (mid)	Net (realistic)	Win rate
14 DTE @ 0.75s	248	\$+103,659	\$+66,459	63%
14 DTE @ 0.60s	251	\$+99,213	\$+61,563	70%
45 DTE @ 0.50s	150	\$+65,508	\$+43,008	71%
45 DTE @ 0.60s	158	\$+65,794	\$+42,094	66%
14 DTE @ 0.50s	258	\$+79,304	\$+40,604	74%
28 DTE @ 0.60s	148	\$+60,688	\$+38,488	70%
28 DTE @ 0.75s	127	\$+45,152	\$+26,102	63%
7 DTE @ 0.75s	267	\$+63,370	\$+23,320	66%

Even the best config (14 DTE wide) is only ~\$40-66k net over 5.5 years after slippage — modest, and it's the ceiling (mid). It is HOLD with very wide wings (near-straddle), heavily reliant on big moves (lumpy). No early-exit or TP variation beats it. This is a thin, non-robust edge, not a convincing standalone strategy.

3. Methodology & the TP correction (data integrity)

verify, don't assume

Why the TP numbers changed (and are now honest):

The engine recorded the TP as the mark-to-market at the first time the profit crossed the target, read from HOURLY quotes. On near-expiry / illiquid legs, those quotes go stale -> the MTM showed PHANTOM peaks (e.g. a TP of \$1160 on a structure whose max gain is the ~\$65 credit). ~18k TP affected.

FIX (validated): the TP keeps the REAL first-cross value (so legitimate gaps/vol-spikes are preserved), but is capped at a generous Black-Scholes plausibility ceiling (vol 3x entry) — only the stale-quote anomalies are removed. 'Keep the real, correct only the anomaly.' 0 phantom TP remain.

Clean throughout: hold = exact intrinsic settle; early exits = time-snapshots (not peak-seeking).

Standard procedure now: physical P&L cap check before any backtest goes to the app.

BOTTOM LINE: even after giving the TP every benefit of the doubt, no IB variation is a convincing standalone edge. Best is wide-wing hold, ~\$40-66k/5.5y realistic — modest, lumpy, mid-optimistic.